

MUKESH NAIDU

ML ENGINEER

Tempe, Arizona | (762) 222-2820 | mnaidu.conn@gmail.com | LinkedIn

OBJECTIVE:

Machine Learning Engineer with 2+ years of experience in designing, developing, and deploying **end-to-end ML solutions** using **Python**, **PyTorch**, and **TensorFlow**. Skilled in data preprocessing, feature engineering, model optimization, and cloud-based deployment on **AWS** and **GCP**. Experienced in building scalable machine learning pipelines, leveraging tools like **Docker**, **FastAPI**, and **Airflow**. Passionate about applying **AI/ML** to solve real-world challenges and continuously exploring the latest advancements in machine learning and deep learning.

KEY ACHIEVEMENTS:

- **Deployed scalable ML pipelines** on AWS SageMaker, streamlining model training workflows and improving efficiency.
- **Enhanced chatbot accuracy** using NLP techniques like BERT and tokenization, leading to a noticeable reduction in fallback rates.
- **Implemented an anomaly detection system** that effectively identified fraudulent transactions in a near real-time environment.
- **Integrated Apache Airflow for ML orchestration**, enabling reliable scheduling of ETL and training tasks.
- **Presented interpretable model insights** using SHAP/LIME to cross-functional teams, supporting data-driven decision-making.

EDUCATION:

Master of Science in Computer Science

University of North Texas, Denton, TX

| May 2024

Bachelor of Technology in Electronics and Communication Engineering

Vellore Institute of Technology, AP, India

| June 2022

TECHNICAL SKILLS:

- **Languages:** Python, SQL, Java
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
- **ML & DS Techniques:** Classification, Regression, Clustering, Hyperparameter Tuning, Model Evaluation, SHAP, LIME
- **Visualization & Databases:** Tableau, Power BI, MySQL, MS Excel
- **Cloud Platforms:** AWS (S3, EC2, SageMaker), GCP (Vertex AI)
- **Deployment Tools:** Docker, FastAPI, CI/CD, REST APIs
- **Version Control & Workflow:** Git, Airflow
- **Data Science & Analytics:** Predictive Modeling, Statistical Analysis, Data Preprocessing, Data Processing

EXPERIENCE:

Machine Learning Engineer | NVIDIA, USA

| Jan 2024 – Present

- Designed, trained, and deployed deep learning models for computer vision and **NLP** using **PyTorch** and **TensorFlow**.
- Built end-to-end **ETL pipelines** with **Apache Spark** and **Pandas**, integrated with **AWS S3** and **Glue**.
- Deployed models via **Docker** and **AWS SageMaker** with **CI/CD integration**, monitored via **CloudWatch**.
- Automated model workflows with **Apache Airflow** for feature extraction and real-time scoring.
- Implemented BERT-based intent classifiers; enhanced chatbot accuracy and reduced fallback rate.
- Used **SHAP** and **LIME** for model explainability and fairness; created executive reports on insights.
- Collaborated cross-functionally in **Agile** teams to deliver scalable **ML solutions** aligned with business goals.

Machine Learning Engineer | Nexova, India

| Jun 2021 – Jul 2022

- Built predictive models **XGBoost**, **scikit-learn** for customer churn and behavior analytics.
- Developed robust preprocessing pipelines: feature engineering, encoding, and anomaly detection.
- Deployed models with **FastAPI**, integrated via **RESTful APIs** with production services.
- Tuned models using **GridSearchCV** and **Optuna**, improved accuracy and F1-score.
- Monitored performance and model drift post-deployment using **real-time alerting systems**.

CERTIFICATIONS:

- [Cloud Machine Learning Engineering and MLOps](#)